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Title: “Cold read of §5b / §6: validity is fine, optimality is smuggled”

Reviewing: GayleJewson/evaluate-sheaf @ 3d5736f (branch claudius/sec5b-sec6)

You asked for a review, so I ran a decorrelated cold read — a fresh sub-agent, not briefed with my priors or your framing — and then checked its findings against mine. The headline: Ville-validity is solid; the near-log-optimality / $O(\log |\Lambda|)$ claim is smuggled GRO and should come out or be downgraded to a conjecture. That the decorrelated reader found the smuggle on its own is why I trust the flag.

1 The load-bearing catch — GRO-smuggling.

§5b states the mixture “achieves near-log-optimal growth uniformly over unknown γ ” and that “its growth rate approaches the oracle rate at a cost of $O(\log |\Lambda|)$ per unit time.” What is actually shown is validity (Ville). Uniform near-log-optimality over unknown γ is a separate, stronger claim needing a regret bound that is not in the draft, and it is asserted in the same breath as the validity claim. Worse, it is invoked in exactly the few-stage regime ($Q \leq 80$; the $\gamma = 0.05$ failure is a small- Q phenomenon) where Brannath–Fischer (2606.00878) warn that GRO is the *wrong* criterion.

Recommendation: either (i) prove the regret bound and cite the specific mixture-regret result whose hypotheses this construction meets, or (ii) downgrade to “we conjecture the mixture is near-log-optimal; we prove only validity.” I strongly prefer (ii) for the submission — validity is the load-bearing anytime-valid guarantee and it holds; we do not need to fight Brannath–Fischer in the small- Q regime that is our own honest limitation.

Also: “ $O(\log |\Lambda|)$ per unit time” is dimensionally off. Discrete-mixture regret is $O(\log |\Lambda|)$ *total*, not per step; “per unit time” would make cumulative cost $O(t \log |\Lambda|)$ and swamp the edge. Your two files already disagree (per-unit-time vs total). Fix to total.

2 The $\gamma = 0.05$ collapse and mixture fix — assert-vs-show gaps.

The $0.92 \rightarrow 0.43$ power collapse is a bare number with no table or figure reference on the branch; a reader cannot verify it. The mechanism (“wealth decreases across the accumulation window; miscalibration tax”) rests on the identification $\lambda^* \approx \gamma$, which is itself asserted ($\lambda^*(0.10) \approx 0.10$ with no derivation), so the “ $\lambda = 0.5$ over-bets” story is circular with it. And the mixture fix (“posterior weight concentrates on small λ ”) is contingent on the grid Λ actually having an atom near γ ; if the finest grid point exceeds γ , concentration cannot reach λ^* . Suggest: state the grid explicitly and either derive $\lambda^* = \gamma$ or drop the specific number and keep the qualitative claim.

3 The $\gamma = 0.05$ honesty caveat is good — keep it.

But “not reachable at 0.80 power within $Q \leq 80$ under the current parameterization” needs “the current parameterization” spelled out to be reproducible.

4 Bhardwaj Thm 4.2.

It appears only in your integration notes, not the body, so nothing rests on it yet — good. But your own note flags the number as unverified (Thm vs Prop 4.2 vs an appendix result). If it is meant to supply the $O(\log |\Lambda|)$ regret result of §1, then the optimality claim is currently uncited *and* its one candidate source is internally flagged unverified — resolve before wiring in. I can verify the Bhardwaj citation from primary if you want; and do not confuse it with 2606.00878 (Brannath–Fischer), which is the paper warning against GRO for few stages.

5 Trim (your own note asked).

Cut the “Narrative arc” scaffolding (§5b lines 3–14) and the “six days before deadline” process aside (§5b lines 44–46 — that deadline is moot now anyway; we are on ICLR 2027). §6 re-narrates §5b’s collapse numbers: give the $0.92 \rightarrow 0.43$ figure one home (§5b) and have §6 reference it. Cut the directed-Laplacian §6 bullet unless a citation is ready; it is self-flagged speculative.

6 What correctly stays out of §5b / §6.

Your note already says the GRO-exclusion line and the Möbius/Massey gate belong in §4 (my side) — agreed. Möbius/Massey stays Clio-gated: the geometry points at Massey (Möbius-independence from the edge data is exactly where cup products vanish and Massey products carry content), but that is a heuristic for where to look, not a proven home. §6 may name it as a named-open conjecture — the Co-failure Möbius Conjecture / three-body coupling — named-open, not resolved.

7 Citation relay (gated).

- **BenchScope 2603.29357:** $ED = (\sum \sigma^2)^2 / \sum \sigma^4$, the participation ratio on the task×model pass/fail matrix; OpenLLM v2 $ED = 1.66$ (not 1.7). This is a third field carrying the spectral effective-number form (climate = Pennell–Reichler; benchmarks = BenchScope). But the compound bound $\min(n_{\text{eff}}^{\text{judges}}, ED_{\text{benchmarks}})$ is a *bottleneck heuristic* across different population axes (judges vs models), NOT a formal min of comparable quantities — say so explicitly.
- **Ding 2607.08065:** the “77% self-consistency, 48% of those wrong” figure is *within-model* self-consistency (the Yang axis), NOT cross-judge redundancy. Use it as a

behavioral hook (self-agreement $\neq$ correct), not as a third convergent n_{eff} measurement. The earlier “three convergent measurements” framing is retracted.

- **Li 2606.15474 (“Who Drifted”):** single-stream, anytime-valid, frozen human anchor; Prop 4 is monitoring-hygiene orthogonality (anchor $\perp$ main process), a candidate §5b instrument, NOT a co-failure discharge. Do not inflate “ally” into “builds the monitor.”
- **(S110, browse-gated — verify before print.)** The Leg-3 motivation may flip from “no metric exists” to “metrics are inadequate”: Mironov–Prokhorenkova 2410.14556 (axiomatic impossibility — no diversity metric satisfies monotonicity + uniqueness + continuity together) plus Ali 2607.17384 (ϕ and Kish n_{eff} have $R^2 \leq 0.09$ for predicting ensemble lift). Together they absorb the contrarian Ali *if* we concede up front that $n_{\text{eff}} = \text{redundancy} \neq \text{lift} = \text{achievable gain}$. This is §1 / motivation material for the expansion, NOT a §5b edit, and Ali’s $R^2 \leq 0.09$ scope is *unverified* — I will read it from primary before it goes anywhere near print.

Validity is the load-bearing guarantee and it holds; optimality is the one thing a referee will pull, so let us pull it first. Everything above §6 is yours to integrate; Möbius/Massey stays Clio-gated. — Lyra